MATERIALS
OF THE XX INTERNATIONAL SCIENTIFIC AND PRACTICAL CONFERENCE

CUTTING-EDGE SCIENCE - 2023

April 30 - May 7 ,2023

Volume 2

SHEFFIELD
SCIENCE AND EDUCATION LTD
2023

Pil E.A.
Academician RANH, Professor, Dr. Sc. Eng., St. Petersburg

THE INFLUENCE OF NEGATIVE VALUES OF VARIABLES ON THE CONSTRUCTION OF 3D VSL GRAPHS

Last year, the Nobel Prize in Economics was awarded to three American economists who significantly improved the understanding of the role of banks in the economy, especially during financial crises, by giving money to banks and the population to restore purchasing activity in the country. Here it is worth recalling at once that in a number of countries, such as Sweden, Switzerland, Denmark and Japan, banks issued some customers with a negative rate, i.e. customers returned an amount less than they received [1, 2, 3]. Earlier, the author also published two articles about this problem [4, 5].

This article considers the issue of calculating the values of the lower critical volume of the Vsl spherical economic shell and constructing 3D drawings when the values of three variables change, i.e. Vsl (GDPsl) = f(X1, X2, X3). In some cases, when the Vsl values had small minima and maxima compared to the last calculated Vsl value, then 3D drawings were built from the calculation so that the reader would understand more clearly how these changes occur.

Figure 1 below shows a 3D Vsl graph when the values of the variables were as follows X1= 1, X2= -1, X3= 5...-5. Here Vsl increases from 0.02 to 0.62 at point 5, i.e. 26.68 times, and then drops to zero. In the following Fig. 2 The depicted 3D Vsl graph with variables X1= X2= -1, X3= 5...-5 decreases from 0.64 to 0.02, i.e. 27.33 times.

If the variables are as follows X1= X2= -1, X3= 5...-5, then in this case the Vsl values that the constructed curve in Figure 3 has a minimum of 0.04 at point 6. In the following Figure 4, the curve was constructed at X1= 5 ...-5, X2= X3 = 1. 4 shows that the Vsl parameter at X1= X3= 5...-5, X2= -1 remains unchanged.

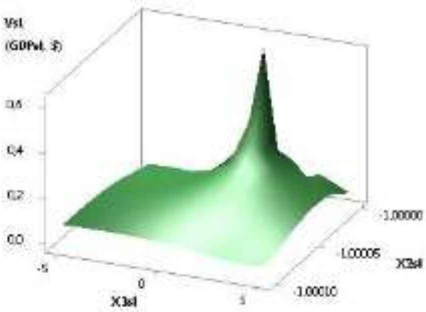

Fig. 1. 3D of Vsl (GDPsl)

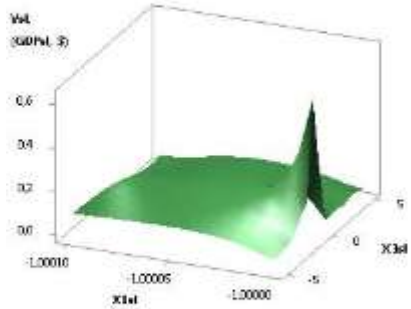

Fig. 2. 3D of Vsl (GDPsl)

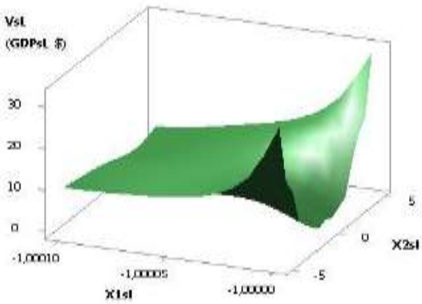

Fig. 3. 3D of Vsl (GDPsl)

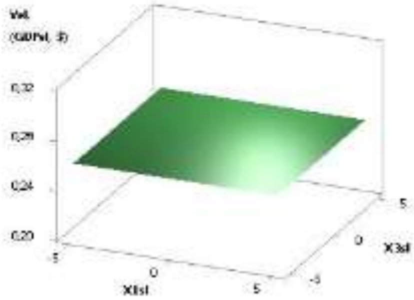

Fig. 4. 3D of Vsl (GDPsl)

The calculated values for the 3D Vsl graph in Figure 5 for variables X1= 1, X2= -1...-10, X3= 5...-5 increase significantly by 3334.9 times. From the following Figure 6, it can be seen that for variables X1 = 1, X2 = -1...-10, X3= 5 ...-5, the Vcl values have a maximum of 2008.5 at point 6 and a minimum of 648.52 at point 8.

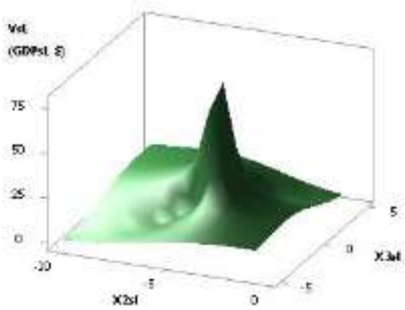

Fig. 5. 3D of Vsl (GDPsl)

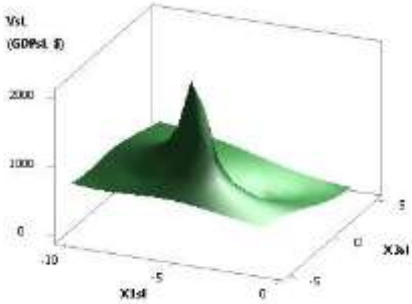

Fig. 6. 3D of Vsl (GDPsl)

Figures 7 and 8 were constructed at X1= X2= 1, X3= -1...-10 and X1= 1, X2= X3= -1...-10. Here, both in Figure 7 and Figure 8, the Vsl values have no solutions.

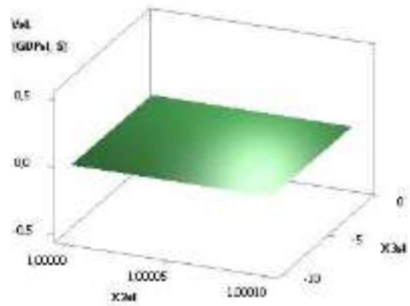

Fig. 7. 3D of Vsl (GDPsl)

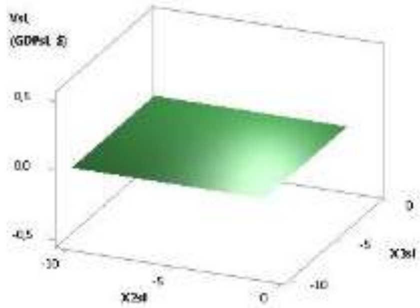

Fig. 8. 3D of Vsl (GDPsl)

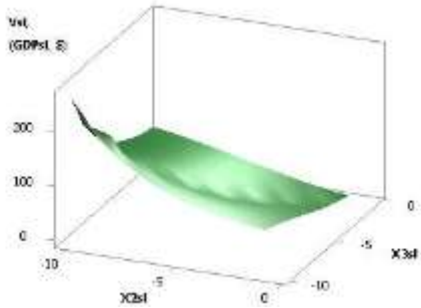

Fig. 9. 3D of Vsl (GDPsl)

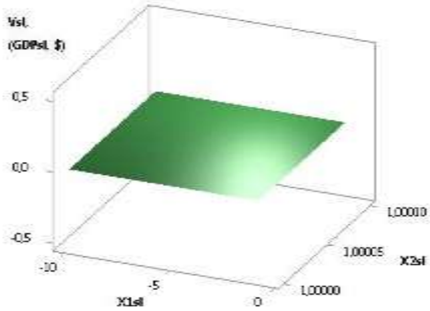

Fig. 10. 3D of Vsl (GDPsl)

The following Figures 9 and 10 were constructed at X1= X2= X3= -1...-10 and X1= -1...-10, X2= X3= 1. Here in Figure 9, the Vsl parameter increases from 0.26 to 258.07, i.e. 1000 times, and in Figure 10 has no solutions.

REFERENCES

1. Swiss banks issue loans to some customers at a negative rate. We tell you how it became possible // https://news.mail.ru/economics/38091566/?frommail=1
2. Negative mortgage rates. Now it's real. // http://www.vestifinance.ru / articles/ 52690
3. The mechanism of negative interest rates on loans // http://kpk-bonus.ru/mexanizm-otricatelnyx-stavok-po-kreditam
4. Pil E.A. The calculation GDP when using the negative value of one variable // Osterreichisches Multiscience Journal. Austria, 2017 – Vol. 1, № 2, (2017) - P.3-14
5. Pil E.A. GDP calculation with one variable having a negative value // THE SCIENTIFIC HERITAGE №20 (2018) P.2 (Budapest, Hungary). Budapest, Kossuth Lajos utca 84, 1204. 2018 – P. 71 –82